Assignment # 1

Artificial

Intelligence

Total Marks: 60 **Submission Date:** 22-03-2026

Name_____**Roll No.** _____

Instructions:

I. Assignment should be hand written.
II. You should create your own solution according to your logic. Any similarity with other student and internet will be considered as cheating and your assignment will be marked as ZERO.
III. Scan your assignment and upload it on portal under submission, your name roll number should be mentioned.

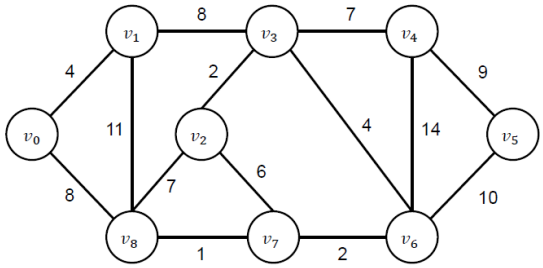
Question # 1: Develop PEAS description of the following task environment [10 Marks]

a. AI-Based Waste Sorting and Recycling System
b. AI-Powered Personalized Learning Tutor
c. Smart Home Energy Optimization System
d. AI-Enabled Wildlife Monitoring and Poaching Prevention System

Question # 2: [10 Marks]

Use the provided graph and apply BFS algorithm w.r.t following table. write out the order in which nodes are added to the explored set, you have to traverse whole graph irrespective of goal state.

Note: keep in view that one node can be explored only once, ignore the edge cost, provide answer using following table method.

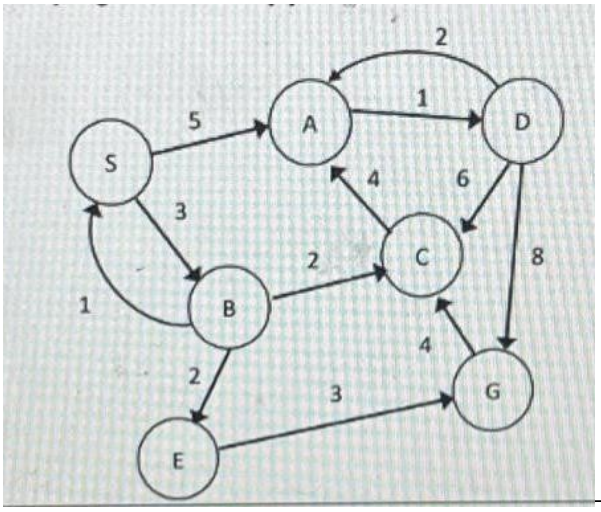


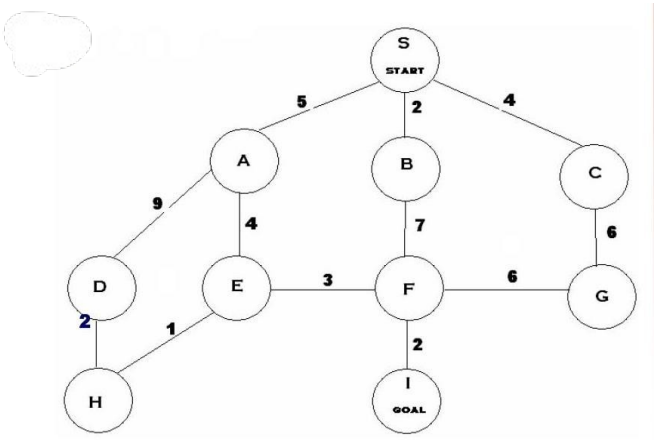
Step	Frontier	Expand[*]	Explored: a set of nodes

Question #3: [40 Marks]

Use the provided graphs and apply **BFS** w.r.t following table on two graphs. write out the order in which nodes are added to the explored set, with **start state S** and **goal state G** in **part a** and **Goal state I in part b**. what is the **returned path?**

Note: Ignore the edge cost





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Step	Frontier	Expand[*]	Explored: a set of nodes